Homework 7: Machine learning on a real prediction problem

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Data: SPY daily closes (dividend-adjusted) from yfinance, 2011-11-03 to 2026-11-02. check_hw07.py passes 5 of 5.

1. The feature table

Six features, all from closes up to and including the row's own day: returns over 1, 5, 20 and 60 days, 20-day realised volatility (annualised), and the close relative to its 50-day average. The label is the next day's close-to-close return. The first 60 days are warm-up and the final day has no label, so 3,770 trading days give 3,709 rows. The hand check rebuilds row 1000 straight from the price series, prints the dates it reads, and asserts it matches build_features exactly.

```
Q1  SPY daily closes 2011-11-03 to 2026-11-02: 3770 trading days
    feature table shape (3709, 7)  (first 60 days are warm-up, last day has no label)
    columns: ['ret_1d', 'ret_5d', 'ret_20d', 'ret_60d', 'vol_20d', 'px_vs_ma50', 'next_day_return']
    first three rows:
                ret_1d    ret_5d  ret_20d  ret_60d  vol_20d  px_vs_ma50  next_day_return
    2012-02-01 0.008757 -0.000679 0.038980 0.055922 0.075342    0.052642         0.001585
    2012-02-02 0.001585  0.006066 0.038997 0.064086 0.075341    0.052392         0.014019
    2012-02-03 0.014019  0.020634 0.050766 0.072337 0.086737    0.064505        -0.000669

    hand check, row 1000 (2016-01-25), close 157.7995:
      ret_1d  = close 2016-01-25 / close 2016-01-22 - 1 = 157.7995 / 160.2216 - 1 = -0.015117
      ret_5d  = close 2016-01-25 / close 2016-01-15 - 1 = 157.7995 / 157.9426 - 1 = -0.000905
      ret_20d = close 2016-01-25 / close 2015-12-23 - 1 = 157.7995 / 173.2566 - 1 = -0.089215
      ret_60d = close 2016-01-25 / close 2015-10-27 - 1 = 157.7995 / 172.7164 - 1 = -0.086366
      vol_20d = stdev of the 20 daily returns 2015-12-24..2016-01-25 x sqrt(252) = 0.205077
      px_vs_ma50 = close / mean(closes 2015-11-11..2016-01-25) - 1 = -0.068319
      label   = close 2016-01-26 / close 2016-01-25 - 1 = +0.013644
    every feature reads closes up to 2016-01-25 and none after;
    only the label reads 2016-01-26. Matches build_features exactly.
```

2. Time-ordered split

60/20/20 by position, no shuffling. The three date ranges do not overlap.

```
Q2  split by time, 60/20/20, no shuffling (dates are the day the features are read)
    training   2012-02-01 to 2020-12-02    2225 rows
    validation 2020-12-03 to 2023-11-14     742 rows
    test       2023-11-15 to 2026-10-30     742 rows
```

3. Four models

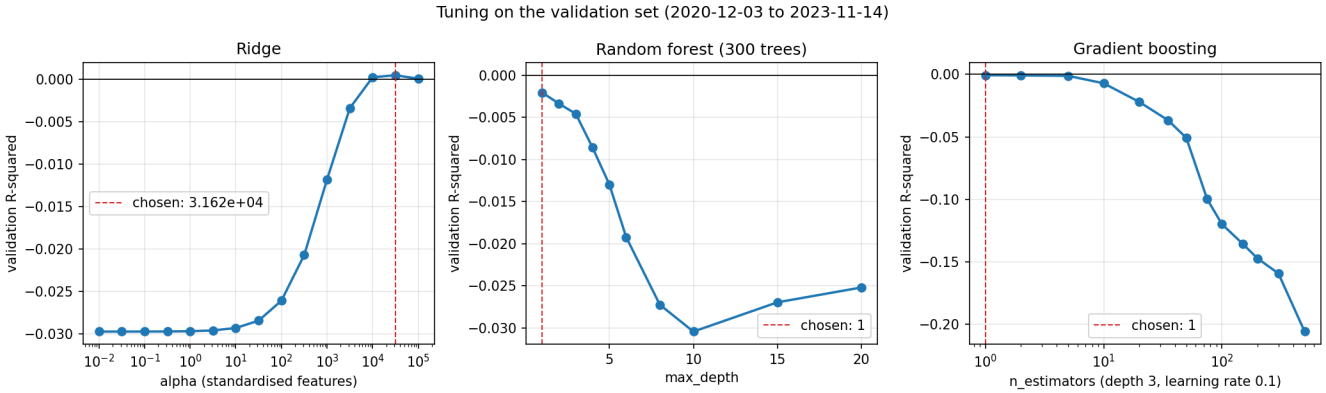
The random forest reaches 0.84 R-squared on training and is negative on validation: it memorised noise. Gradient boosting does the same at 0.45. All four are negative on validation, meaning they predict worse than a constant equal to the validation mean.

```
Q3  R-squared predicting next-day return
    model                training  validation
    linear regression      0.0287     -0.0298
    ridge (alpha=1)        0.0287     -0.0297
    random forest          0.8424     -0.0371
    gradient boosting      0.4549     -0.1199
```

4. Tuning on validation

Ridge: alpha (features standardised inside the pipeline). Random forest: max_depth. Gradient boosting: number of trees. Every curve climbs toward the most constrained end: ridge shrinks almost to zero, the forest wants stumps, boosting wants one tree. On this validation set the best model is one that barely moves away from predicting the training average.

```
Q4  one hyperparameter each, chosen on validation R-squared
    ridge             alpha = 3.162e+04 validation R2 0.0005
    random forest     max_depth = 1     validation R2 -0.0021
    gradient boosting n_estimators = 1  validation R2 -0.0007
```



5. Best model, once, on test

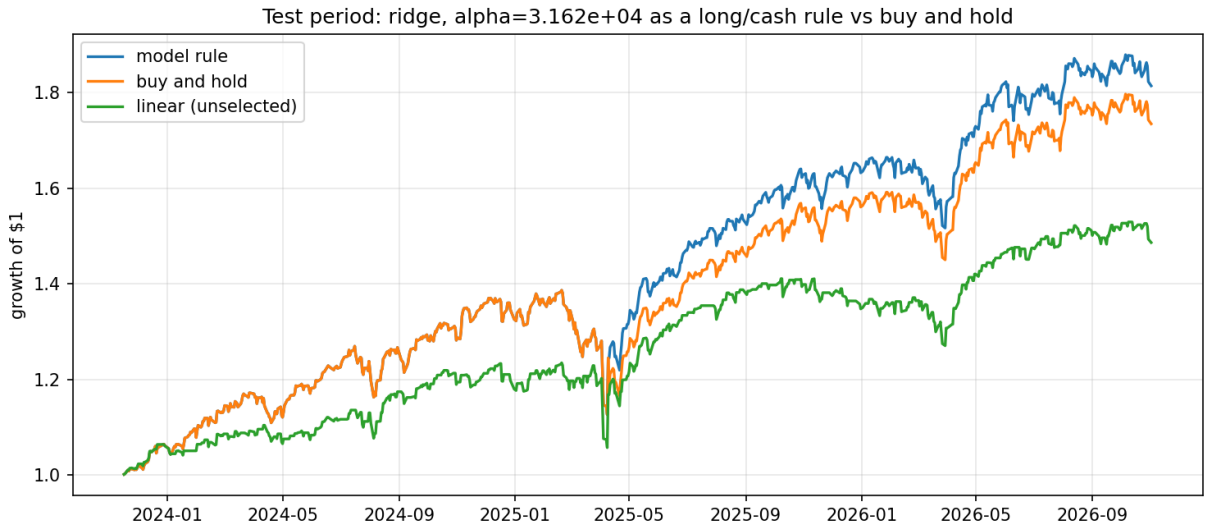
Ridge at alpha = 31,623 has the best validation R-squared (0.0005, essentially zero). Its predictions range from -0.02% to +0.17% a day and are positive on all but one test day. Directional accuracy (56.6%) is the same as the share of up days (56.5%), the baseline from up_day_baseline.

```
Q5  best on validation: ridge, alpha=3.162e+04 (validation R2 0.0005)
    test set, run once: R2 0.0026   corr(predicted, actual) +0.1043
    predicted up on 99.9% of test days; directional accuracy 56.6% vs 56.5% of test days actually up
```

6. Trading rule backtest

This uses the Homework 4 backtester convention: on day i the rule sees the closes ending with today's and, if it holds, earns the return from close i to close i+1. It is handed the 60 closes before the test period so the window is full on the first test day. Buy and hold run through the same backtester matches the raw price ratio, and the window-based rule matches the feature-table predictions day for day (both asserted in the code). The unregularised linear model is shown for comparison only. It was not the one selected.

```
Q6  hold SPY when the prediction is positive, cash otherwise; test period 2023-11-15 to 2026-11-02
                        annualised  max drawdown  invested    total
    model rule              22.41%       -18.76%     99.9%    81.4%
    buy and hold            20.56%       -18.76%    100.0%    73.4%
    linear (unselected)     14.40%       -14.37%     63.1%    48.6%
    model rule in cash on 2025-04-09 (that day +10.50%), missing 2025-04-10 (-4.38%)
```



The chosen rule beats buy and hold by about 8 points in total, but all of that comes from one day: after the +10.5% rally on April 9, 2025, the large negative coefficient on the 1-day return pushed the forecast below zero, and the rule sat out the -4.4% on April 10. One day is an anecdote, not evidence.

7. What the models relied on

Coefficients are per one-standard-deviation move in each feature, in basis points of next-day return. Every model puts the 1-day return first, with a negative sign in the linear models. That is short-term mean reversion, which the April 2025 example shows. Tree importances come from the tuned (very shallow) trees.

```
Q7  linear regression on standardised features: coefficient per 1 s.d. move, ranked by size
    ret_1d        -17.28 bp
    ret_60d        -5.73 bp
    vol_20d        +3.20 bp
    px_vs_ma50     +3.01 bp
    ret_5d         +2.50 bp
    ret_20d        -1.07 bp
    chosen ridge (alpha=3.162e+04), same scale:
    ret_1d         -1.05 bp
    ret_60d        -0.42 bp
    px_vs_ma50     -0.36 bp
    ret_5d         -0.27 bp
    vol_20d        +0.27 bp
    ret_20d        -0.26 bp
    random forest (max_depth=1) importances, ranked:
    ret_1d         0.503
    ret_60d        0.240
    px_vs_ma50     0.130
    ret_5d         0.087
    ret_20d        0.033
    vol_20d        0.007
    gradient boosting (n_estimators=1) importances, ranked:
    ret_1d         0.580
    px_vs_ma50     0.226
    ret_5d         0.194
    ret_20d        0.000
    ret_60d        0.000
    vol_20d        0.000
```

8. Written

Why a validation R-squared near zero is the expected result, not a bug. SPY is the most heavily traded security in the world, and these six features are computed from its public price history in a few lines of code. Thousands of funds, high-frequency firms and students have run this exact regression, with more data, faster execution and lower costs than I have. If yesterday's return or the distance from the 50-day average reliably predicted tomorrow, they would trade it today, and that trading would push today's price until the predictable part was gone. What is left in next-day returns is almost all noise. The scale makes the point. The average daily return over these 15 years is 0.059% and the daily standard deviation is 1.04%. Even knowing the true mean exactly would explain only about 0.3% of the variance (0.059 squared over 1.04 squared), so the most a genuine edge could show is an R-squared in the thousandths. The flexible models do worse than zero because they fit the noise: the forest's 0.84 training R-squared is memorisation, and it costs them out of sample. The regime also shifts: training is 2012 to 2020 and validation includes the 2022 bear market, so even the training average misses the validation average. Tuning confirmed all of this, because each model's best setting was the one closest to predicting a constant. A bug would show up the other way, as a validation R-squared well above zero, which usually means leakage (shuffling, or a feature that reads ahead). The checker and the hand check rule that out here.

Can the trading rule still make money with near-zero R-squared? Yes. R-squared scores the squared size of every forecast error on every day, compared with always guessing the mean. A long/cash rule only uses the sign of the forecast, and its P&L is weighted by the size of the move on the days it acts. Those are different targets, for three reasons.

First, drift: stocks go up on average, so a model that just says "up" nearly every day has an R-squared of about zero and still earns the equity premium. That is what my chosen rule did: invested 99.9% of the time, 22.4% a year. It made money because the market went up, not because of skill.

Second, R-squared punishes scale and correlation does not. On test the ridge forecasts had a 0.10 correlation with actual returns, yet R-squared was only 0.003, because the heavy shrinkage made the forecasts tiny. Rescaling them would raise R-squared to about corr squared (0.011), but the signs, which are all the rule uses, would not change at all. In practice a correlation of 0.05 to 0.10 (the information coefficient) is a usable signal when it is applied across many independent bets, because the edge compounds with breadth.

Third, what matters is being right on the big days. A rule that is wrong slightly more often on small days but avoids a few large losses can earn more than buy and hold. Mine beat it by about 8 points by missing a single -4.4% day.

The honest caveats: that excess return rests on one day; dropping late March to April 2025 lowers the correlation to about 0.07. The rank correlation is only 0.06, the validation correlation was 0.03, and transaction costs and taxes are ignored. The unselected linear rule, which traded much more, earned less (14.4% a year) but stayed invested only 63% of the time and had a smaller drawdown (-14.4% vs -18.8%). So a near-zero R-squared does not rule out a profitable rule. But this test cannot tell my result apart from luck, and "made money" is not the same as "has an edge over buy and hold."